

Reference Designs Of

Title of Reference Design	Reference Design of a Smart Plug Featuring Solid-State Relay	Reference Design of a Smart Plug With Grid Frequency Monitoring
Image		
Key Attractions	The reference design employs a solid-state relay for switching on/off power and utilises Wi-Fi connectivity to enable users to control the plug using a smartphone, tablet, computer, etc.	The smart plug reference design can enable IoT capabilities for any household appliance running at a load up to 10A. It can allow the user to control the appliance and monitor power consumption along with setting the switch on/off time for the future.
Highlights	TIDC-SMARTPLUG-WIFI is a reference design for a smart plug with Wi-Fi connectivity. This design implements single-outlet energy measurement with remote connect/disconnect capability. The reference design is based on the MSP430i2040 microcontroller unit (MCU). The smart plug uses SimpleLink Wi-Fi to connect from any smartphone, tablet, or computer through a standard web browser. The reference design can enable designers to quickly create networked load control devices for industrial building and home automation applications. It features a solid-state relay that offers good isolation and enables the user to remotely connect/disconnect the power to the device. The circuit of this reference design includes voltage input circuit, current input circuit, and design considerations of Wi-Fi system-on-chip (SoC). The reference design features low-power components plus efficient power supply, which provides low system power consumption. The TIDC-SMARTPLUG-WIFI reference design can also be used for single-phase energy measurement that calculates RMS current, RMS voltage, active and reactive power and energies, power factor, and frequency. The system design is tested and includes firmware for energy measurement, Wi-Fi connectivity, and relay control along with Android based demo application and users' guide.	This smart plug reference design can measure single-phase power and enable wireless control of equipment. The NXP smart plug reference design enables the user to add smart functionality to regular appliances. It enables the user to control or monitor energy usage of home appliances drawing a maximum current of 10 amps. The reference design is based on the MKM34Z64 microcontroller. The smart plug reference design can be extended to various areas, including lighting monitor, machine room power distribution, etc. The design implements measurement of grid voltage, current, frequency, active and reactive energies. With the Wi-Fi chip, it uses UART and predefined command protocol to implement status display and function setting from the mobile application to the socket. This reference design can be adjusted for different loads by changing the values of relay and width of power PCB. The NXP smart plug reference design also offers features such as socket on/off status display, and the user can view current active and reactive power, apparent power, and power consumption history in the application provided by the company.
Application	The module is suitable for switching on and off a device or for monitoring power consumption of any household appliance.	The NXP smart plug is suitable for regular home appliances, such as fans, lights, television, etc.
OEM Brand	Texas Instruments	NXP
URL	https://www.electronicsforu.com/electronics-projects/reference-design-for-wi-fi-enabled-smart-plug	https://www.electronicsforu.com/electronics-projects/industry-grade-projects/reference-design-for-a-smart-plug-that-can-measure-1%ct%95-power